

JUPITER ONBOARDING
SC26 TUTORIAL *SESSION 1B*

15 November 2026 | Andreas Herten | Jülich Supercomputing Centre, Forschungszentrum Jülich

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Accessing JUPITER

- Everything listed on GitHub repo of tutorial:
  https://go.fzj.de/mg-gh[1]

[1] Unshortened link: https://github.com/FZJ-JSC/tutorial-multi-gpu/

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  `https://go.fzj.de/mg-gh`[1]

1. Create JSC account at JuDoor
2. Join `training2660` project

→ `https://go.fzj.de/mg-jd`

3. Accept usage agreement
4. Wait 15 minutes 
5. Access system via Jupyter 4.5
   JUPITER, *`training2660`*, ***`LoginNode`***

→ `https://go.fzj.de/mg-jup`

6. Source course environment in a Jupyter Shell
   ```
   $ source $PROJECT_training2660/env.sh
   ```
7. Gather course material
   ```
   $ jsc-material-sync
   ```

[1] Unshortened link: `https://github.com/FZJ-JSC/tutorial-multi-gpu/`

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- Please start process now
- We'll repeat the following steps in the first hands-on session

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- SC26 Program

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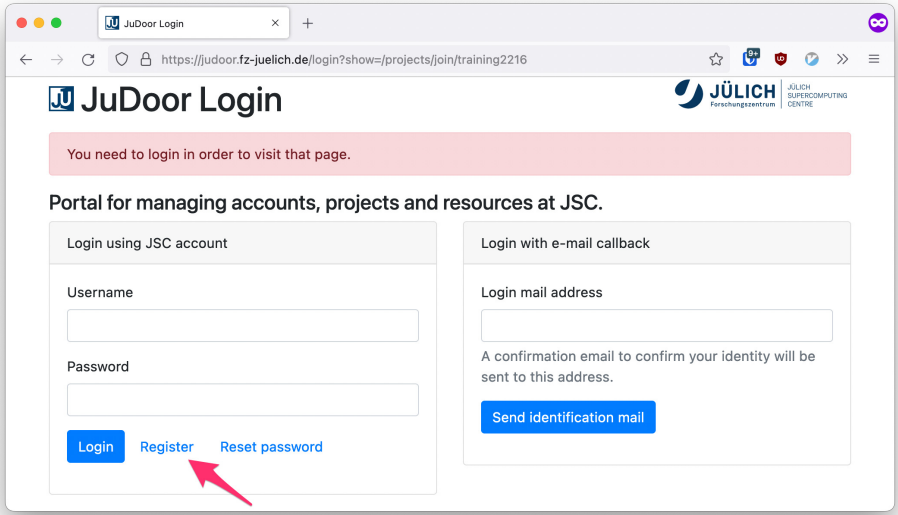
   ```
   $ source $PROJECT_training2660/env.sh
   ```

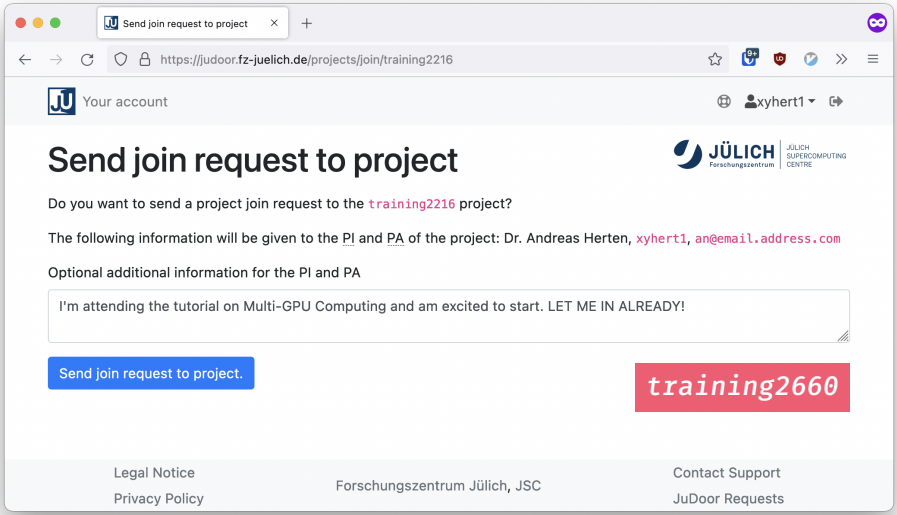
7. Gather course material

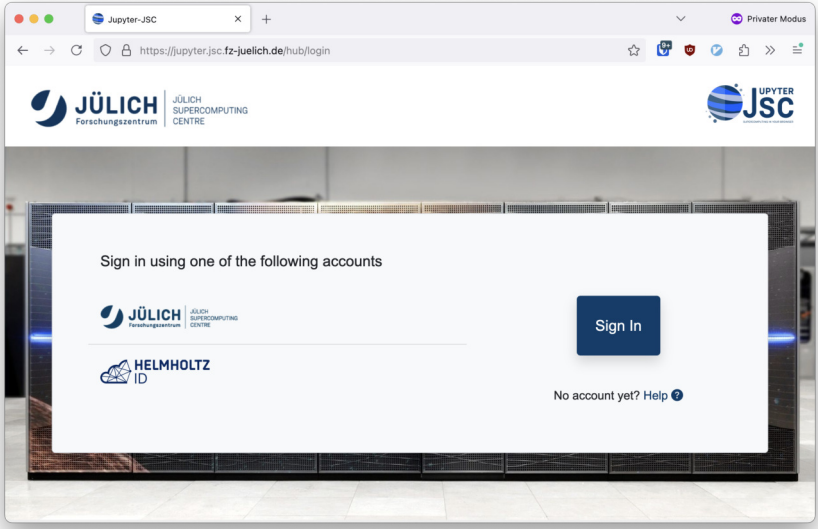
   ```
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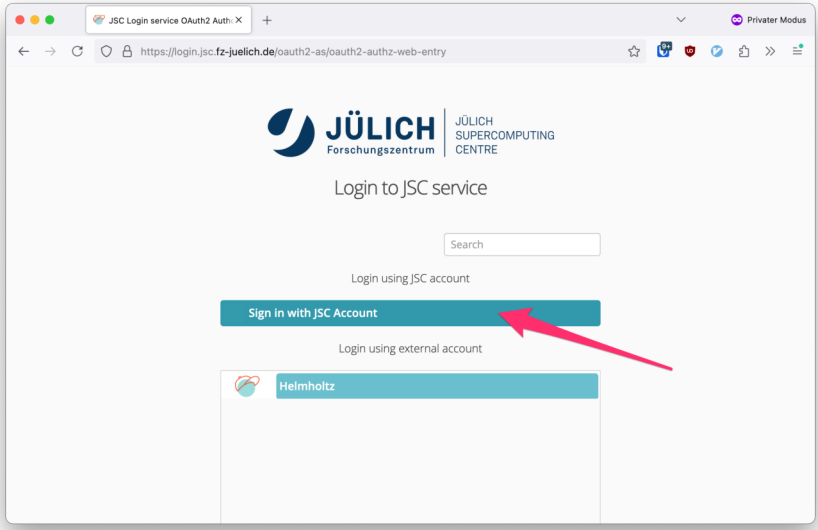
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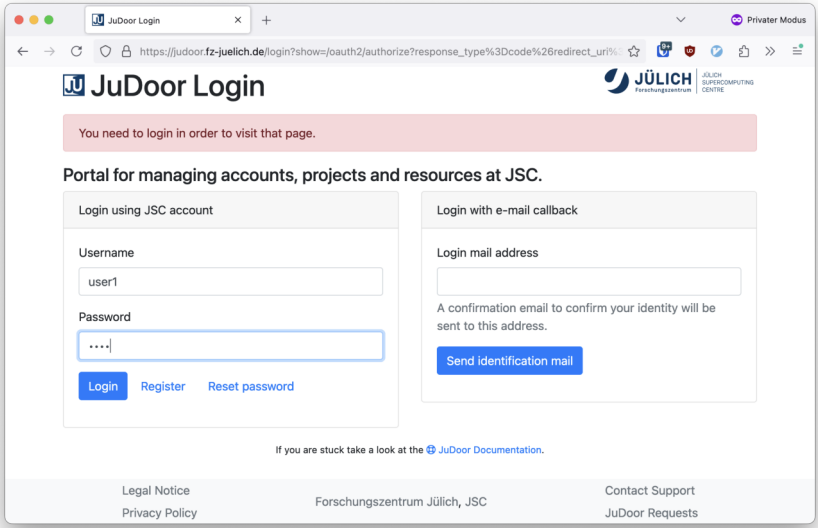
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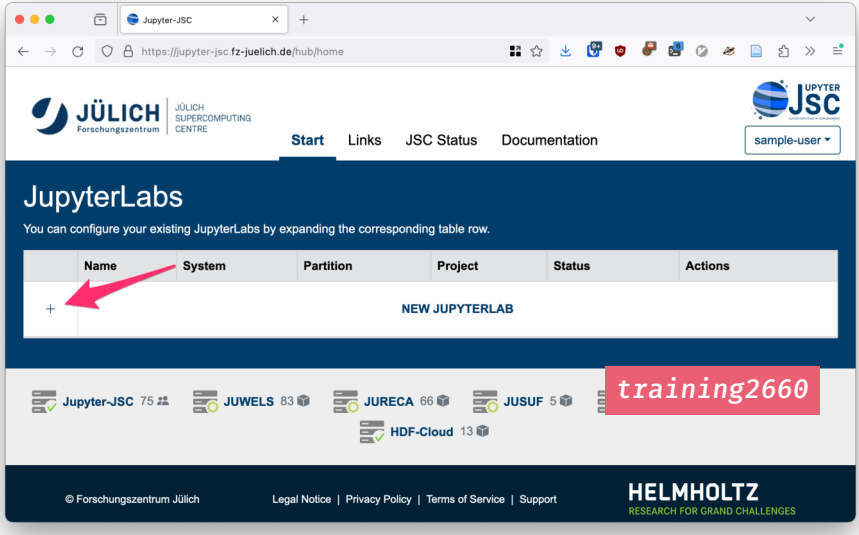

JuDoor Login
https://judoor.fz-juelich.de/login?show=/projects/join/training2216
JuDoor Login
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JÜLICH SUPERCOMPUTING CENTRE
You need to login in order to visit that page.
Portal for managing accounts, projects and resources at JSC.
Login using JSC account
Username
Password
Login
Register
Reset password
Login with e-mail callback
Login mail address
A confirmation email to confirm your identity will be sent to this address.
Send identification mail

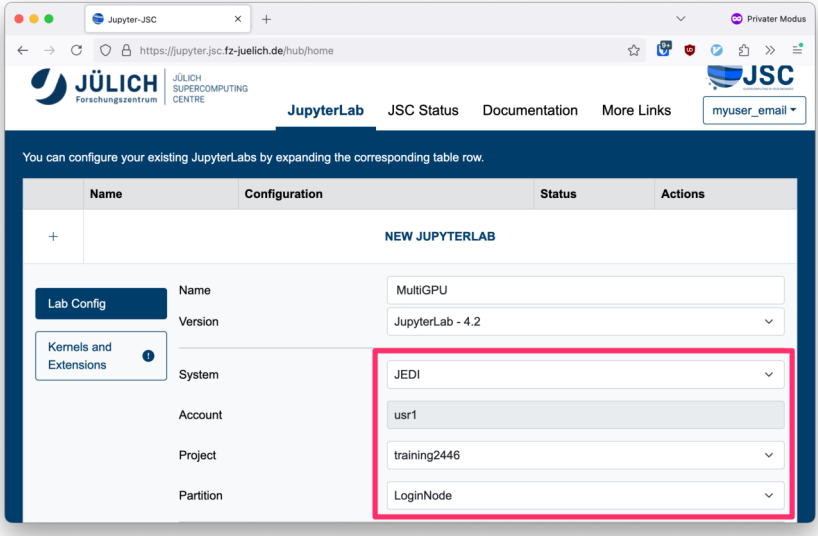

Send join request to project
https://judoor.fz-juelich.de/projects/join/training2216
Your account
xyhert1
Send join request to project
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JÜLICH SUPERCOMPUTING CENTRE
Do you want to send a project join request to the training2216 project?
The following information will be given to the PI and PA of the project: Dr. Andreas Herten, xyhert1, an@email.address.com
Optional additional information for the PI and PA
I'm attending the tutorial on Multi-GPU Computing and am excited to start. LET ME IN ALREADY!
Send join request to project.
training2660
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Contact Support
JuDoor Requests

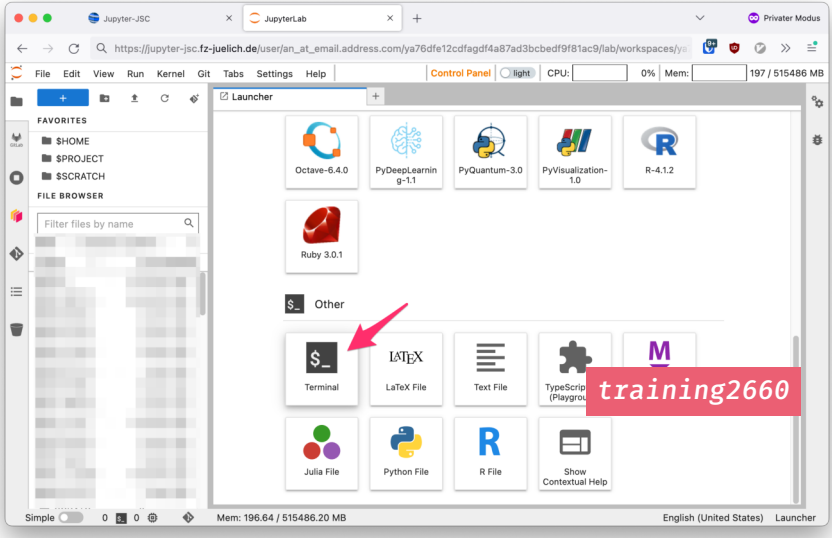

Jupyter-JSC
Privater Modus
https://jupyter.jsc.fz-juelich.de/hub/login
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SUPERCOMPUTING
CENTRE
UPYTER
JSC
Sign in using one of the following accounts
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Forschungszentrum
JÜLICH
SUPERCOMPUTING
CENTRE
HELMHOLTZ
ID
Sign In
No account yet? Help

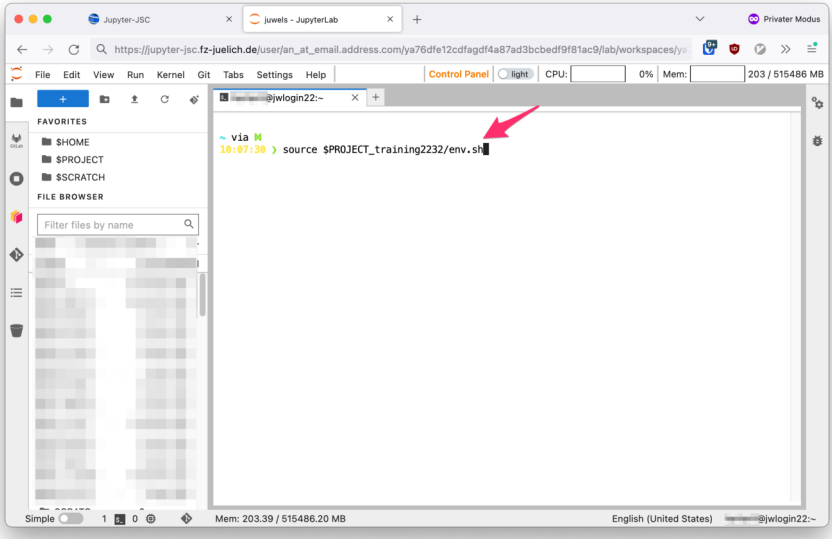

JSC Login service OAuth2 Auth
Privater Modus
https://login.jsc.fz-juelich.de/oauth2-as/oauth2-authz-web-entry
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SUPERCOMPUTING
CENTRE
Login to JSC service
Search
Login using JSC account
Sign in with JSC Account
Login using external account
Helmholtz

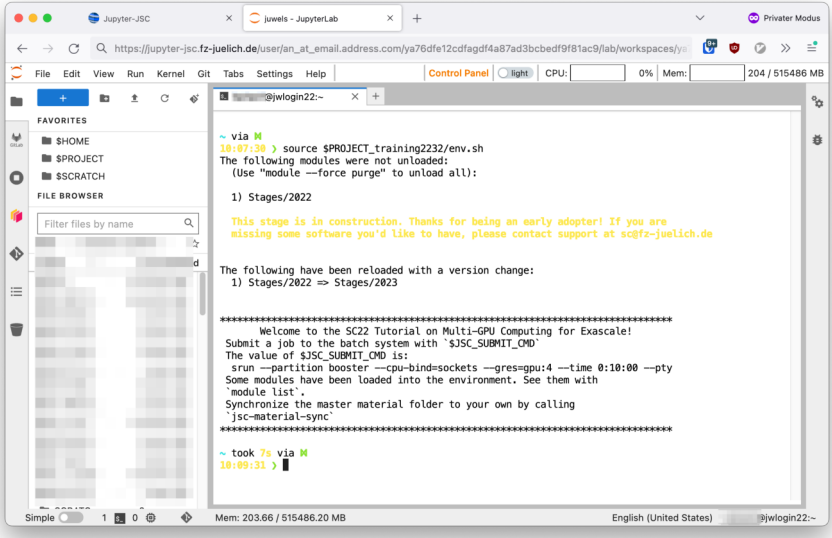

JuDoor Login
Privater Modus
https://judoor.fz-juelich.de/login?show=/oauth2/authorize?response_type%3Dcode%26redirect_uri%
JuDoor Login
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JÜLICH SUPERCOMPUTING CENTRE
You need to login in order to visit that page.
Portal for managing accounts, projects and resources at JSC.
Login using JSC account
Username
user1
Password
Login
Register
Reset password
Login with e-mail callback
Login mail address
A confirmation email to confirm your identity will be sent to this address.
Send identification mail
If you are stuck take a look at the JuDoor Documentation.
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Contact Support
JuDoor Requests


Jupyter-JSC
https://jupyter-jsc.fz-juelich.de/hub/home
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JUPYTER JSC
Start
Links
JSC Status
Documentation
sample-user
JupyterLabs
You can configure your existing JupyterLabs by expanding the corresponding table row.
Name
System
Partition
Project
Status
Actions
+
NEW JUPYTERLAB
Jupyter-JSC 75
JUWELS 83
JURECA 66
JUSUF 5
HDF-Cloud 13
training2660
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RESEARCH FOR GRAND CHALLENGES


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JSC
JupyterLab
JSC Status
Documentation
More Links
myuser_email
You can configure your existing JupyterLabs by expanding the corresponding table row.
Name
Configuration
Status
Actions
NEW JUPYTERLAB
Lab Config
Kernels and Extensions
Name
MultiGPU
Version
JupyterLab - 4.2
System
JEDI
Account
usr1
Project
training2446
Partition
LoginNode


Jupyter-JSC
JupyterLab
Privater Modus
https://jupyter-jsc.fz-juelich.de/user/an_at_email.address.com/ya76dfe12cdfagdf4a87ad3bcbedf9f81ac9/lab/workspaces/ya
File Edit View Run Kernel Git Tabs Settings Help
Control Panel
light
CPU:
0%
Mem:
197 / 515486 MB
FAVORITES
$HOME
$PROJECT
$SCRATCH
FILE BROWSER
Filter files by name
Launcher
Octave-6.4.0
PyDeepLearning-1.1
PyQuantum-3.0
PyVisualization-1.0
R-4.1.2
Ruby 3.0.1
Other
Terminal
LaTeX File
Text File
Julia File
Python File
R File
Show Contextual Help
training2660
Simple
Mem: 196.64 / 515486.20 MB
English (United States)
Launcher


Jupyter-JSC
juwels - JupyterLab
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https://jupyter-jsc.fz-juelich.de/user/an_at_email.address.com/ya76dfe12cdfagdf4a87ad3bcbedf9f81ac9/lab/workspaces/ya
File Edit View Run Kernel Git Tabs Settings Help
Control Panel
light
CPU:
0%
Mem:
203 / 515486 MB
@jwlogin22:~
FAVORITES
$HOME
$PROJECT
$SCRATCH
FILE BROWSER
Filter files by name
~ via M
10:07:30 ❯ source $PROJECT_training2232/env.sh
Simple
1
0
Mem: 203.39 / 515486.20 MB
English (United States)
@jwlogin22:~


Jupyter-JSC
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https://jupyter-jsc.fz-juelich.de/user/an_at_email.address.com/ya76dfe12cdfagdf4a87ad3bcbedf9f81ac9/lab/workspaces/ya
File Edit View Run Kernel Git Tabs Settings Help
Control Panel
light
CPU:
0%
Mem:
204 / 515486 MB
FAVORITES
$HOME
$PROJECT
$SCRATCH
FILE BROWSER
Filter files by name
@jwlogin22:~
~ via M
10:07:30 ❯ source $PROJECT_training2232/env.sh
The following modules were not unloaded:
(Use "module --force purge" to unload all):
1) Stages/2022
This stage is in construction. Thanks for being an early adopter! If you are
missing some software you'd like to have, please contact support at sc@fz-juelich.de
The following have been reloaded with a version change:
1) Stages/2022 => Stages/2023
Welcome to the SC22 Tutorial on Multi-GPU Computing for Exascale!
Submit a job to the batch system with `$JSC_SUBMIT_CMD`
The value of $JSC_SUBMIT_CMD is:
srun --partition booster --cpu-bind=sockets --gres=gpu:4 --time 0:10:00 --pty
Some modules have been loaded into the environment. See them with
`module list`.
Synchronize the master material folder to your own by calling
`jsc-material-sync`
~ took 7s via M
10:09:31 ❯
Simple
Mem: 203.66 / 515486.20 MB
English (United States)
@jwlogin22:~

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Profiling Tools

- Extra Credits: Prepare for *Profiling Session*
- Download **Nsight Systems** now; install!

→ `https://developer.nvidia.com/nsight-systems/get-started`

- Also: Via package manager `developer.download.nvidia.com/devtools/repos`

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SSH Login

SSH Login

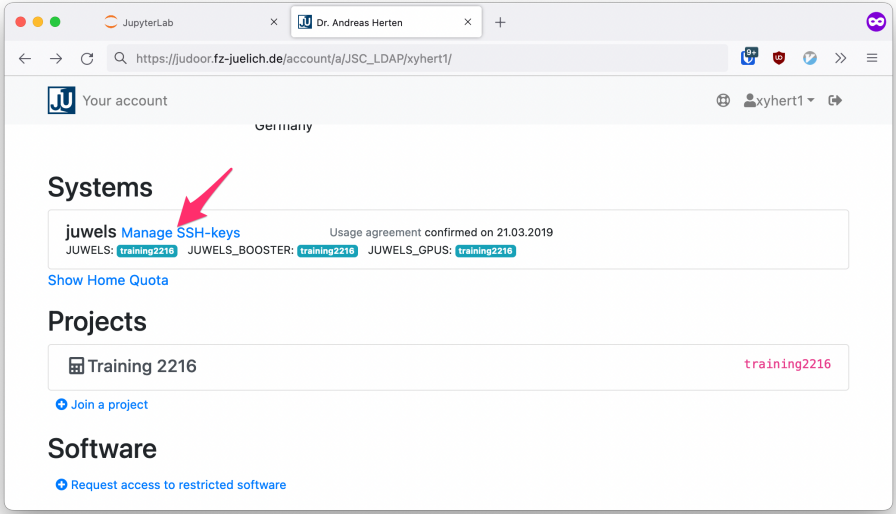
- Login with SSH available
- We recommend Jupyter JSC: easier, more features
- Add SSH key via JuDoor to JUWELS Booster
- Important: `from` clause (limits connections to be from defined sources)
- Example

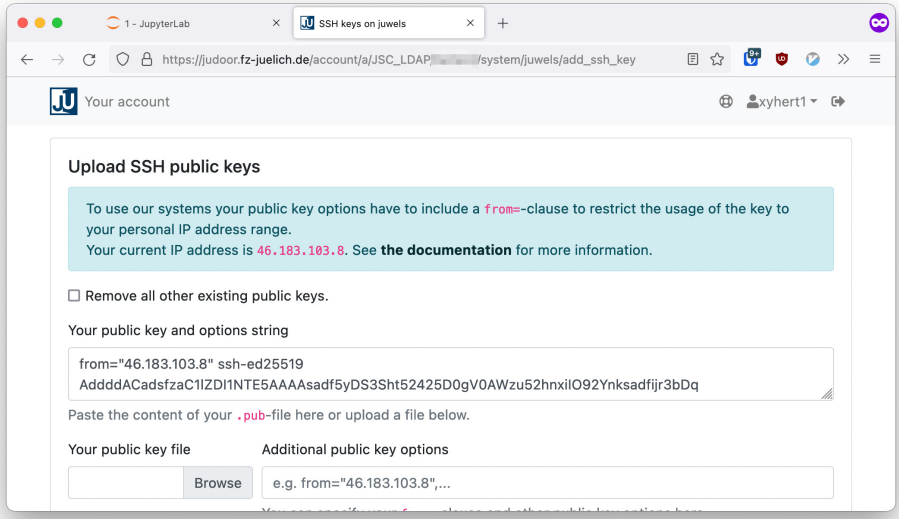
```
from="80.146.183.0/24" ssh-ed25519 AddddACadsfzaC1lZDI1NTE5AAAAsa
# coarser: from="80.144.0.0/13"
```

→ SSH: `ssh user1@login.jupiter.fz-juelich.de`

- Help at `apps.fz-juelich.de/jsc/hps/jupiter/access.html`

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JupyterLab
Dr. Andreas Herten
https://judoor.fz-juelich.de/account/a/JSC_LDAP/xyhert1/
Your account
xyhert1
Germany
Systems
juwels Manage SSH-keys
Usage agreement confirmed on 21.03.2019
JUWELS: training2216 JUWELS_BOOSTER: training2216 JUWELS_GPUS: training2216
Show Home Quota
Projects
Training 2216
training2216
Join a project
Software
Request access to restricted software


1 - JupyterLab
SSH keys on juwels
https://judoor.fz-juelich.de/account/a/JSC_LDAP[redacted]/system/juwels/add_ssh_key
Your account
xyhert1
Upload SSH public keys
To use our systems your public key options have to include a from=-clause to restrict the usage of the key to your personal IP address range.
Your current IP address is 46.183.103.8. See the documentation for more information.
Remove all other existing public keys.
Your public key and options string
from="46.183.103.8" ssh-ed25519
AddddACadsfzaC1lZDI1NTE5AAAAsadf5yDS3Sht52425D0gV0AWzu52hnxilO92Ynksadfijr3bDq
Paste the content of your .pub-file here or upload a file below.
Your public key file
Browse
Additional public key options
e.g. from="46.183.103.8",...

QR Codes



GitHub repo:

`https://go.fzj.de/mg-gh`



JuDoor:

`https://go.fzj.de/mg-jd`



Jupyter Portal:

`https://go.fzj.de/mg-jup`

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